Flask Web Application Implementation

Import Required Libraries

The required libraries are imported to build the web application. **Flask** is used as the web framework to create and manage application routes, **Pickle** is used to load the trained Human Development Index (HDI) prediction model stored in a **.pkl** file, while **NumPy** and **Pandas** are used to process numerical input data and organize it into a suitable format for prediction.

Initialize the Flask Application

The Flask application is initialized using the **Flask()** class. The trained HDI prediction model is loaded from the Pickle file using **pickle.load()**, allowing the application to generate predictions without retraining the model each time it is executed.

Define Application Routes

The **'/'** route renders the **home.html** page, which introduces the HDI prediction system. The **'/Prediction'** route displays the input form (**indexnew.html**) where users enter feature values. The **'/Home'** route allows users to return to the home page.

Prediction Route

The **'/predict'** route accepts **POST** requests, retrieves user-entered values from the form, converts them into floating-point numbers, stores them in a Pandas DataFrame, and passes them to the trained machine learning model. The predicted HDI score is categorized as **Low HDI**, **Medium HDI**, **High HDI**, or **Very High HDI** based on predefined ranges. Finally, the prediction result is displayed on the **resultnew.html** page.

```python

# importing the necessary dependencies
import numpy as np #used for numerical analysis
import pandas as pd # used for data manipulation
from flask import Flask, render_template, request
# Flask-It is our framework which we are going to use to run/serve our
#request-for accessing file which was uploaded by the user on our appl
import pickle
app = Flask(__name__) # initializing a flask app
model = pickle.load(open('HDI.pkl', 'rb')) #loading the model

@app.route('/')# route to display the home page
def home():
    return render_template('home.html') #rendering the home page
```

```python
@app.route('/')# route to display the home page
def home():
    return render_template('home.html') #rendering the home page
@app.route('/Prediction',methods=['POST','GET'])
def prediction():
    return render_template('indexnew.html')
@app.route('/Home',methods=['POST','GET'])
def my_home():
    return render_template('home.html')

@app.route('/predict',methods=['POST'])# route to show the predictions in a web UI
def predict():

```

```python
def predict():

    #reading the inputs given by the user
    input_features = [float(x) for x in request.form.values()]
    features_value = [np.array(input_features)]

    features_name = ['Country','Life expectancy','Mean years of schooling','Gross national income (GNI) per

    df = pd.DataFrame(features_value, columns=features_name)

    # predictions using the loaded model file
    output = model.predict(df)
    print(round(output[0][0],2))
    print(type(output))
    y_pred =round(output[0][0],2)
    if(y_pred >= 0.3 and y_pred <= 0.4) :
        return render_template("resultnew.html",prediction_text = 'Low HDI'+ str(y_pred))
    elif(y_pred >= 0.4 and y_pred <= 0.7) :
        return render_template("resultnew.html",prediction_text = 'Medium HDI '+str(y_pred))
    elif(y_pred >= 0.7 and y_pred <= 0.8) :
        return render_template("resultnew.html",prediction_text = 'High HDI'+str(y_pred))
    elif(y_pred >= 0.8 and y_pred <= 0.94) :
        return render_template("resultnew.html",prediction_text = 'Very High HDI'+str(y_pred))
    else :
        return render_template("resultnew.html",prediction_text = 'The given values do not match the range of values


    # showing the prediction results in a UI# showing the prediction results in a UI
    return render_template('resultnew.html', prediction_text=output)


if __name__ == '__main__':
    # running the app
    app.run(debug=True,port=5000)
```